

Unitree Week 11+12

Presentation by Mario Ortiz

This week

- Studying how to simulate/train the Unitree humanoid robots to do tasks with perception.
 - I learned that the current simulations I worked on with Isaac Lab & Isaac Sim utilize **Proprioceptive Blind Locomotion**, meaning they do not see and utilize touch to detect things such as staircase.
 - Will research methods to implement sensors such as LiDAR onto Isaac Lab so the G1 can analyze terrains with perception.

More advanced tasks for the G1

- Reinforcement learning on its own may not be the most effective method for doing more complicated tasks such as viewing and picking up an item. Therefore it may be more beneficial to train the G1 using a pipeline of different methods.
 - “Teleoperation → Mimic Learning (BC/Diffusion) → Reinforcement fine-tuning → Evaluation”

Teleoperation

- First we must collect data to begin training. This is done with Teleoperation, where the human controls the robot in real-time with Isaac Sim.
- We then collect joint angles, sensor data, etc. from this human control, which we can then implicate to training.

Mimic/Imitation Learning

- Mimic Learning (or Imitation Learning) would be the initial training phase in which the robot learns to replicate the actions from humans via teleoperation.
- Two types of Mimic Learning are:
 - Behavioral Cloning: Simpler technique that looks at images of the actions, predicts the motor commands the human gave to initiate the action.
 - Diffusion Policy: Difficult but more efficient technique that generates a set of possible actions and from them chooses a high precision trajectory.

Reinforcement Learning

- We can now use RL for optimizing the training results from Mimic Learning.
- We can create a reward function, rewarding the robot for completing the task, else penalizing the robot if it doesn't.

Imitation Learning using Video Data

- Video-Based Skill Transfer: extracts object trajectories and human movement from a single video, creating a “trajectory prior” of waypoints for the robot to follow, which is then refined through residual training.”
- Latent Video Planning: “integrates visual diffusion models into robot policies, allowing for real-time (approx. 14 Hz) generation of actions and future, time-aligned video frames from multiple views, significantly reducing reliance on extensive robot data.”

Task-Aware Digital Cousin

- **Task-Aware Digital Cousin** is simulated video data that closely emulates a specific task.
 - Pros:
 - can align simulation with a real-world
 - Can automate data generation
 - Cons:
 - Difficult to generate simulations that emulate real-world tasks without error. Hence, lot's of tuning must be done.



Digital Cousin Workflow

1. Scene Generation: turning a real-world photo into a fully interactive simulation (no manual 3D modeling)
 - Per-object Extraction: identifies every object in the real-world image
 - Digital Asset Matching: Using a library, we find a functionally similar 3D model (the cousin).
 - Spatial Grounding: Calculate where the objects are in 3D space and spawn them in to the environment matching the simulated layout (can use Isaac Sim).
2. Task-Aware Simulation Setup: Setting up training → the simulation is setup to complete the specific task that occurs in the photo/video.
 - Load the R1/G1 (URDF or USD file for example) and setup constraints
3. Imitation learning setup:
 - Seed Demonstration: provide demos (Isaac Lab's teleoperation capabilities)
 - Data Augmentation: using the demos, calculate the same movements for the G1/R1 robot's joints.
 - Policy Training
 - Integrate Dataset onto the real G1.

Real-to-sim generation

Turning real-world photos/videos into a simulated environment:

- Grounding DINO → object detection
- DINOv2 → object description; generates feature embeddings for each object to describe “visual essence”
- Segment Anything (SAM 2) → generates pixel-masks of objects detected by DINO
- Objaverse/PartNet-Mobility → 3D model databases containing assets for the “cousin” that matches the DINOv2 features.
- Depth-Anything-v2 → Estimates the distance and depth of objects in the photo
- NVIDIA NuRec → Helps turn photo data into a USD scene (integrated with Isaac Lab; easily can match with the Unitree G1’s USD file).

Simulation Setup + Imitation Learning Setup

- Isaac Sim: Simulator
 - Unitree G1/R1 .usd file
 - Isaac Lab Manager API: used for “task-aware” logic (the joint, randomization, reward parameters)
 - cuRobo: GPU-accelerated library that ensures the robot’s movements are physically possible.
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- AnyTrack/WHAM: maps the entity in the video (human/R1) to the G1’s joints.
 - Isaac Lab Teleop: records control inputs into an HDF5 file format; for training and in the future for movement of the robot.
 - SkillGen/MimicGen: Creates the thousands of “cousin” demos from the seed demos (~10 seed demos for example).
 - Robomimic: Training framework; uses the HDF5 dataset to train the neural network
 - Unitree SDK/CycloneDDS: Send the trained policy commands to the PC then to the physical robot.

Real-to-sim generation

Directory: C:\Users\Mario\IsaacLab\source\isaacsim\mimic

Mode	LastWriteTime	Length	Name
d-----	2/12/2026 10:24 AM		config
d-----	2/12/2026 10:24 AM		docs
d-----	2/12/2026 10:47 AM		isaacsim_mimic
d-----	2/12/2026 12:55 PM		isaacsim_mimic.egg-info
d-----	2/12/2026 10:24 AM		test
-a----	2/12/2026 10:24 AM	101	pyproject.toml
-a----	2/12/2026 10:24 AM	2177	setup.py

Directory: C:\Users\Mario\IsaacLab\scripts\imitation_learning

Mode	LastWriteTime	Length	Name
d-----	2/12/2026 10:24 AM		isaacsim_mimic
d-----	2/12/2026 10:24 AM		locomanipulation_sdg
d-----	2/12/2026 10:24 AM		robomimic

Downloaded Curobo

```
git clone https://github.com/NVlabs/curobo.git
```

```
cd curobo
```

```
C:\Users\Mario\IsaacLab\_isaac_sim\python.bat -m pip install -e . --no-build-isolation
```

```
Obtaining file:///C:/Users/Mario/IsaacLab
```

Sources

- https://isaac-sim.github.io/IsaacLab/main/source/overview/core-concepts/sensors/ray_caster.html
- <https://github.com/isaac-sim/IsaacLab/issues/4033>
- <https://www.youtube.com/watch?v=JfdPxD2vmIY>
- <https://www.youtube.com/watch?v=zIWcqZCyE-s>
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